

MA2031-Hints to solve problems of Assignment V

1. Define T^* so that $\langle x, \alpha \rangle = \langle x, T^* \alpha \rangle$ is satisfied. $T^* \alpha = \alpha u$
2. Define T^* so that $\langle x, y \rangle = \langle x, T^* y \rangle$ is satisfied.
3. T is invertible iff $\exists S$ such that $ST = TS = I$. Prove using the definition of invertibility and adjoint.
4. If W is a subspace of an inner product space V , we have proved in assignment 3 that $W + W^\perp = V$.
5. Trivial.
6. Fix a non-zero vector $z \in \ker(f)^\perp$, show that any vector $x \in V$ can be written as $x = w + kz, w \in \ker(f), k \in \mathbb{F}$, verify that $y = \frac{f(z)}{\langle z, z \rangle} z$ is the required vector
7. Show that the $N(A)$ is non-trivial, show that any non-zero vector from $N(A)$ satisfies the requirements
8. (a) If $Ax = 0$ for $x \neq 0$, say $x = (x_1, x_2, \dots, x_n)$, then try to find a non-zero linear combinations of columns of A that gives zero vector.
 (b) Show that the solution space is a non-trivial subspace of R^n
 (c) Produce suitable examples, for instance, take a matrix which satisfies (a) or (b), For example carefully add rows of the matrix so that the conditions are satisfied is one way of doing it.
 (d) Consequence of (b)
 (e) Prove the contrapositive of the statement is the columns of A are linearly independent iff $Ax = 0$ has only finitely many solutions (i.e., trivial solution)
9. (a) Use the fact that $Ax = b$ has solution iff $b \in R(A)$
 (b) Use the fact that $Ax = b$ has solution iff $b \in R(A)$ and if it has more than one solution, arrive the contradiction $\text{rank}(A) < n$
 (c) Use the fact that $Ax = b$ has solution iff $b \in R(A)$ and generate infinitely many solutions from the non-trivial null space of A
 (d) Follows directly from the fact that $Ax = b$ has solution iff $b \in R(A)$
 (Note: $R(A) = \text{span}\{a^1, a^2, \dots, a^n\}$)
 (e) Use the fact that the columns of A are linearly independent iff $N(A) = 0$, use the above fact to prove that if $b \in R(A)$ we have unique solution. If $b \notin R(A)$ there is nothing to prove.
10. $\dim(U) = k \cdot \dim(N(A))$, Observe that each column of the vector (matrix) in U has $\dim(N(A))$ free variables, and hence the result follows.

11. Solve the system of equations $A^T Ax = A^T b$
12. If $A = QR$, Justify the reason for i th the columns of Q is $q^i = \pm g^i$, where g^i is the i th Gram Schmidt vector, when the process is applied to the columns of A . Also use the positivity of R to show the uniqueness of Q